

Effects of *Lepidium sativum*, *Nigella sativa* and *Trigonella foenum-graceum* on Phenytoin Pharmacokinetics in Beagle Dogs

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The present work was designed to evaluate the effect of some commonly used herbs viz. garden cress (*Lepidium sativum*), black seed (*Nigella sativa*) and fenugreek (*Trigonella foenum-graceum*) on the disposition of phenytoin after oral administration in a dog model. Phenytoin was given orally at a dose of 50 mg, and blood samples were obtained for the determination of drug's pharmacokinetic parameters. After a suitable washout period, animals were commenced on a specific herb treatment for one week. Garden cress treatment caused a modest increase in maximum observed concentration (C_{max}) and terminal half-life ($T_{1/2\lambda}$) of phenytoin with a reduction in clearance by 33%. The effect of black seed therapy was more drastic on drug elimination and to a lesser extent on its volume of distribution at steady state (V_{ss}) with a consequent reduction in systemic exposure measured by area under the curve ($AUC_{0-\infty}$) by about 87%. The effect of fenugreek therapy resembled, albeit to a lesser extent, that of black seed with a significant reduction in $AUC_{0-\infty}$ by ~72%. In addition, there was a 73% increase in V_{ss} . Our findings suggest that the phenytoin disposition can be significantly altered by the concurrent consumption of tested herbal products. Copyright © 2013 John Wiley & Sons, Ltd.

Keywords: herbal interaction; *Lepidium sativum*; *Nigella sativa*; phenytoin; pharmacokinetics; *Trigonella foenum-graceum*.

INTRODUCTION

Herbal remedies contain more than one active constituent such as alkaloids, flavonoids, glycosides, anthraquinones, saponins and others, which have the potential to interact with conventional drugs through different mechanisms. It has also been demonstrated that many herbal products undergo metabolism catalyzed by Cytochrome P450 (CYP) and to elicit different effects on CYP-mediated pathways including induction and inhibition (Fang *et al.*, 2011; Han *et al.*, 2011; Ho *et al.*, 2011; Doehmer and Eisenbraun, 2011). For example, Ginkgo (*Ginkgo biloba*) has been shown to interact with many drugs. One study showed that Ginkgo's flavonoids increase the activity of CYP3A4, which is responsible for the metabolism of the antidepressant drug trazodone to an active metabolite. This is a clinically significant interaction as some patients with Alzheimer's disease fell into coma after taking a combination of trazodone and Ginkgo (Izzo and Ernst, 2001).

Several herbal remedies are particularly common in many Asian countries for a variety of health-related problems. For example, the seeds of *Nigella sativa*, commonly known as black seed or black cumin, are used in folk medicine for the treatment and prevention of a number of conditions which include asthma, diarrhea and dyslipidemia, hypertension, diabetes, inflammation,

colic, headache, eczema, fever and influenza (Muhtasib *et al.*, 2006). The important ingredients in the seed include thymoquinone (TQ), nigellone, isoquinoline alkaloids and fixed oils (El-Dakhkhny, 1963; Ali and Blunden, 2003). TQ has been reported to block lipopolysaccharide-induced phosphorylation of p38, mitogen-activated protein kinase, extracellular-regulated kinases and nuclear factor-kappaB-p65 (Vaillancourt *et al.*, 2011). Recent studies also showed that the *Nigella sativa* oil may be considered as a potential adjuvant therapy in rheumatoid arthritis (Gheita and Kenawy, 2012).

Fenugreek, another commonly used herbal product, is the dried ripe seed of *Trigonella foenum-graecum*. The constituents of fenugreek are simple alkaloids consisting mainly of trigonelline, choline, gentianine, carpaine and saponine (Balandrin *et al.*, 1985). It is used as nutrient and in treating kidney ailments, diabetes, cellulites and tuberculosis. Fenugreek has also been used to reduce plasma cholesterol level and enhance anti-coagulant activity (Sowmya and Rajyalakshmi, 1999; Basch *et al.*, 2003). Enhancement of male libido has also been seen with the standardized fenugreek and its mineral formulation (Steels *et al.*, 2011). Garden cress is the dried ripe seed of *Lepidium sativum*. The major constituents of garden cress seed are glucosinolates. Other constituents include ascorbic acid, cucurbitacins and cardenolides (Gokavi *et al.*, 2004; Kassie *et al.*, 2003). The herb is used for respiratory disorders, vitamin C deficiency, constipation, poor immunity and as a diuretic (Nadkarni, 1954). The practitioners of Indian medicine consider the seeds are useful in dysenteric diarrhea as well as in febrile and catarrhal infections. The antidiarrheal

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and antispasmodic activities of garden cress are species dependent, mediating its antispasmodic effect through combinations of multiple pathways including activation of $K^{(+)}$ channels, and inhibition of muscarinic receptors, $Ca^{(++)}$ channels and phosphodiesterase enzyme (Gilani *et al.*, 2012; Rehman *et al.*, 2012). The seeds are also considered to be lactagogue and are also administered postpartum after boiling with milk (Vohora and Khan, 1977; De Smet 1997).

Given the fact that self-medication is prevalent practice worldwide, information about possible drug–herb interactions is of a critical importance. Easterford and colleagues (Easterford *et al.*, 2005) carried out a study on 400 epileptic patients and reported that 34% patients had used or were using complementary and alternative medicines for general health purposes considering these medicines to have fewer or no effect on their epilepsy. Therefore, the interactive properties of commonly used herbs in the Asian cultures with clinically used drugs require further investigation. The effects of garden cress, black seed and fenugreek on the disposition of phenytoin, a well-know anticonvulsant drug with a narrow therapeutic index, have been explored with the current study.

MATERIALS AND METHODS

Chemicals and materials. Black seed (Voucher # 15966), garden cress (Voucher # 15967) and fenugreek (Voucher # 15968) dated December 1st 2010 were purchased from a local market and authenticated by a Taxonomist at the College of Pharmacy, King Saud University, Riyadh, Saudi Arabia. Phenytoin, butobarbital and potassium dihydrogen phosphate were purchased from Sigma-Aldrich Co. (St. Louis, MO, USA). Acetonitrile, and methanol, of HPLC grade were obtained from BDH Laboratory (BDH Chemicals Ltd., Poole, UK). All other reagents and chemicals used in the assay were of the highest purity available for analytical research; water was of a Milli-Q quality.

Animals and study protocol. Fifteen healthy male beagle dogs weighing 15–20 kg were obtained from the College of Pharmacy, Animal Care and Use Center, King Saud University, Riyadh, Saudi Arabia, and were divided into three groups (five in each group). Animals were maintained in accordance with the recommendations of the 'Guide for the Care and Use of Laboratory Animals' approved by the center were used in the study (Clearance No. 021003-0301; January 15, 2011). Animals were allowed free water and food *ad libitum* during the study except that the chow was pulled 12 h prior to animal dosing. After the overnight fasting, dogs were given a 30 g of dog food to serve as controls. Fifteen minutes later, a 50 mg phenytoin tablet was administered orally to each dog. Blood samples (1 mL) were collected from leg vein through an indwelling vein catheter 18 G into heparinized vacutainer tubes before phenytoin administration and at 0.25, 0.5, 0.75, 1, 1.5, 2, 3, 4, 6, 8, 12, 24, 36, 48, 60 and 72 h post dosing. Equal volumes of normal saline were injected through the cannula to replace the fluid loss. Plasma was immediately separated by centrifugation of blood at 2500 g for 10 min and stored at -20°C pending analysis.

After a suitable washout period (minimum of 7 days), animals per group were started on a garden cress regimen at a dose of 7.5 g, black seed at a dose of 2.5 g or fenugreek at a dose of 25 g with 30 g of dog food twice daily for seven days. On the morning of the last day and after an overnight fasting, the last garden cress, black seed or fenugreek dose was given; phenytoin (50 mg orally) was then administered 15 min later. The same sampling scheme was repeated as described above to determine drug plasma concentrations and to evaluate the effect of herb therapy on phenytoin disposition.

Phenytoin assay. Aliquots of 200 μL plasma containing phenytoin were pipetted into 1.5 mL Eppendorf tubes, and 25 μL internal standard (butobarbital) was added and vortexed for 10 s. To extract phenytoin, 1.0 mL of diethyl ether was added to each tube and vortexed to mix for 1.0 min and centrifuged at 3500 g for 5.0 min. The supernatant was transferred to another Eppendorf tube and evaporated to dryness under a gentle stream of nitrogen. The evaporate residue was reconstituted with 200 μL mobile phase, and 50 μL of it was injected into the HPLC system. The HPLC system consisted of a Waters model 717 autosampler, model M-600 quaternary HPLC pump, an inline degasser and model M-2487 dual UV absorbance detector set at $\lambda=230\text{ nm}$. The mobile phase consisted of potassium dihydrogen phosphate (10 mM), methanol and acetonitrile in the ratio of 64:33:3 per volume, which was delivered at a flow rate of 1.5 mL/min through a stainless steel Nova-pack C18, $4.6 \times 75\text{ mm}$, 5-micron column. Signal output was captured using Millennium³² software, version 3.05 (Waters Corp., Milford, MA, USA).

A calibration curve was prepared with each batch of test samples by spiking known amounts of phenytoin and the internal standard to blank dog plasma and treated in the same manner as described above. Standard curves for phenytoin were linear over the range of 0.25–10 $\mu\text{g/mL}$. The mean coefficient of determination (r^2) for the standard curves was >0.99 . The calculated intra- and inter-day CVs for phenytoin QC samples were less than 6% with greater than 97% accuracy.

Pharmacokinetic analysis. Non-compartmental analysis was used to determine the pharmacokinetic parameters of phenytoin in the beagle dogs before and after herbal treatment. The maximum observed plasma concentration (C_{max}) and the time to reach the maximum observed plasma concentration (T_{max}) following phenytoin oral administration were determined from the collected data. The area under the concentration *versus* time curve (AUC_{0-t}) was calculated by the trapezoidal rule with extrapolation to infinity. AUC from time 0 to infinity ($\text{AUC}_{0-\infty}$) was calculated as the sum of AUC_{0-t} and any residual area, which was computed as the concentration at the last time point divided by terminal rate constant. The phenytoin apparent elimination rate constant (λ_z) was estimated by linear regression analysis of the terminal portion of the log concentration–time data; apparent elimination half-life ($T_{1/2\lambda}$) was calculated as $\ln 2/\lambda$. Total body clearance (CL) was computed as $\text{dose}/\text{AUC}_{0-\infty}$ and volume of distribution at steady state (V_{ss}) as $(\text{AUMC}_{0-\infty}/\text{AUC}_{0-\infty}) \times \text{CL}$. Pharmacokinetic analysis was performed using WinNonlin software (version 4.1, Pharsight Corporation, Palo Alto, CA, USA).

Statistical analysis. Data are presented as the mean±S.E.M. Differences in pharmacokinetic parameters of phenytoin before and after treatment with studied herbs were assessed by a Student's *t*-test on log-transformed data. Statistical significance will be assumed when $p \leq 0.05$. All calculations will be performed using GraphPad Prism version 3.00 for Windows (San Diego, CA, USA).

RESULTS

Plasma concentration–time curve and pharmacokinetic parameters of phenytoin in dogs following an oral administration before and after garden cress are given in Fig. 1 and Table 1. Both $C_{\max}$ and $AUC_{0-\infty}$ were increased by 16% and 49%, respectively, as compared to controls (Table 1; $p < 0.01$). The calculated oral clearance (CL/F) decreased by 33% (0.72 L/h vs. 0.48 L/h; $p < 0.001$), while the estimated apparent oral volume of distribution at steady state (V_{ss}/F) was reduced by about 20% (22.9 L vs. 18.4 L; $p = 0.0047$). Changes in these parameters resulted in a modest increase in $T_{1/2\lambda}$ (~26 h vs. ~22 h; $p = 0.0918$).

Treatment with black seed caused more drastic effects on phenytoin disposition. The plasma concentrations and estimated non-compartmental pharmacokinetic

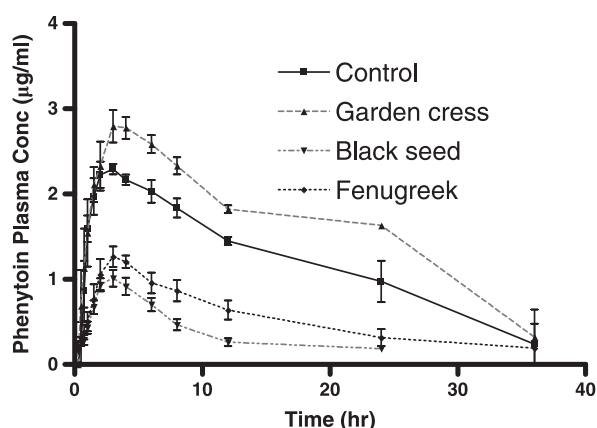


Figure 1. Plasma–time concentration curve of phenytoin following oral administration of 50 mg dose to male beagle dogs ($n = 5$) and showing the effect of garden cress, black seed and fenugreek treatment on phenytoin disposition *in vivo*.

parameters of the anticonvulsant drug before and after herbal treatment are shown in Fig. 1 and Table 2. Similarly, $C_{\max}$ and $AUC_{0-\infty}$ were statistically decreased after black seed treatment as opposed to controls ($p < 0.0001$). There was an increase in CL/F from 0.72 L/h to 5.8 L/h ($p < 0.0001$), as well as a two-fold increment in V_{ss} (22.9 L vs. 45.9 L). This has led to a significant reduction in phenytoin half-life to about 5 h ($p < 0.001$).

Plasma concentration–time curve of phenytoin in dogs following an oral administration before and after fenugreek consumption is illustrated in Fig. 1. Similar to that effect of black seed, drug clearance was significantly increased (~285%), and both $C_{\max}$ and $AUC_{0-\infty}$ were reduced by 44% and 72%, respectively ($P < 0.0001$); the estimated V_{ss}/F was also increased from 22.9 L to 39.6 L ($p < 0.0001$). Consequently, phenytoin's $T_{1/2\lambda}$ was reduced to about 10 h compared to 22 h in controls (Table 3).

DISCUSSION

Most herbs contain more than one active constituent such as alkaloids, glycosides, saponins, coumarins, anthraquinones and flavonoids, which have the potential to interact with conventional drugs through different mechanisms (Wills *et al.*, 2000). Therefore, the multitude of those constituents apparently increases the risk of drug interactions. This renders any analysis of these interactions a complex and difficult task (Cupp, 1999). To best of our knowledge, this work is the first designed to evaluate the effect of garden cress, black seed and fenugreek on the disposition of phenytoin after oral administration in an animal model.

Many herbal remedies have been shown to have interactions with drugs via CYP450. These interactions are of more concern particularly for drugs that have a narrow therapeutic index. Izzo and Ernst reviewed interactions between herbal products and prescribed drugs (Izzo and Ernst, 2001). For example, St John's wort (*Hypericum perforatum*) lowers blood concentrations of cyclosporin, amitriptyline, digoxin, indinavir, warfarin, phenprocoumon and theophylline. Furthermore, it causes intermenstrual bleeding, delirium or mild serotonin syndrome when used concurrently with oral contraceptives, loperamide or selective serotonin-reuptake inhibitors (e.g. sertraline, paroxetine,

Table 1. Non-compartmental pharmacokinetic parameters of phenytoin following an oral administration in beagle dogs before and after garden cress treatment.

Parameter	Estimate (Mean±SEM)		<i>P</i> value
	Before	After	
$C_{\max}$ ($\mu\text{g mL}^{-1}$)	2.478±0.06361	2.894±0.1282	0.0167
$T_{\max}$ (h)	2.500±0.3162	3.800±0.5831	0.0675
AUC_{0-t} ($\mu\text{g}\cdot\text{h min}^{-1}$)	35.51±2.155	47.34±1.671	0.0045
$AUC_{0-\infty}$ ($\mu\text{g}\cdot\text{h min}^{-1}$)	70.4155±4.3978	104.1167±3.0755	<0.001
CL/F (mL h^{-1})	722.9741±51.9149	481.8778±13.9545	<0.001
V_{ss} (mL)	22893±574.1	18438±914.1	0.0047
λ_z (h^{-1})	0.0324±0.0016	0.0275±0.002	0.0925
$T_{1/2\lambda}$ (h)	21.6145±1.0443	25.7444±1.8684	0.0918
MRT (h)	32.0741±1.5234	38.5551±2.8236	0.0816

Table 2. Non-compartmental pharmacokinetic parameters of phenytoin following an oral administration in beagle dogs before and after black seed treatment.

Parameter	Estimate (Mean±SEM)		P value
	Before	After	
C _{max} (µg mL ⁻¹)	2.478±0.06361	1.075±0.0773	<0.0001
T _{max} (h)	2.500±0.3162	2.8±0.3742	0.5713
AUC _{0-t} (µg*h min ⁻¹)	35.51±2.155	7.7613±1.1357	<0.0001
AUC _{0-∞} (µg*h min ⁻¹)	70.4155±4.3978	9.2346±1.2988	<0.0001
CL/F (mL h ⁻¹)	722.9741±51.9149	5764.5589±633.8129	<0.0001
V _{ss} (mL)	22893±574.1	45954±4550	0.0001
λ _z (h ⁻¹)	0.0324±0.0016	0.1559±0.0227	<0.001
T _{1/2λ} (h)	21.6145±1.0443	4.9997±0.9707	<0.001
MRT (h)	32.0741±1.5234	8.403±1.3258	<0.0001

Table 3. Non-compartmental pharmacokinetic parameters of phenytoin following an oral administration in beagle dogs before and after fenugreek treatment.

Parameter	Estimate (Mean±SEM)		P value
	Before	After	
C _{max} (µg mL ⁻¹)	2.478±0.06361	1.3541±0.0846	<0.0001
T _{max} (h)	2.500±0.3162	3.8±1.0677	0.2510
AUC _{0-t} (µg*h min ⁻¹)	35.51±2.155	16.8912±2.5256	0.0010
AUC _{0-∞} (µg*h min ⁻¹)	70.4155±4.3978	19.7993±2.451	<0.0001
CL/F (mL h ⁻¹)	722.9741±51.9149	2781.6217±371.9541	<0.0001
V _{ss} (mL)	22893±574.1	39621±6688	0.0116
λ _z (h ⁻¹)	0.0324±0.0016	0.075±0.0085	<0.001
T _{1/2λ} (h)	21.6145±1.0443	9.6541±0.9389	<0.001
MRT (h)	32.0741±1.5234	14.1256±0.6764	<0.0001

nefazodone), respectively. Ginseng (*Panax ginseng*) lowers blood concentrations of warfarin and induces mania if used concomitantly with phenelzine. Garlic (*Allium sativum*) changes pharmacokinetic parameters of paracetamol, decreases blood concentrations of warfarin and produces hypoglycemia when taken with chlorpropamide. Kava (*Piper methysticum*) increases 'off' periods in Parkinson patients taking levodopa and can cause a semicomatose state when given concomitantly with alprazolam. Finally, phenytoin has been reported to have increased clearance and frequent seizures occurred when combined with Ayuevedic syrup (Patsalos and Perucca, 2003). Interestingly, fenugreek and its constituents including trigonelline, gentianine, histidine, lysine, arginine and quercetin may potentiate the effects of CYP2D6 and CYP3A4, and also interfere with the cytotoxic effects of cyclophosphamide (Lambert and Cormier, 2001; Izzo *et al.*, 2005). *Nigella sativa* and its principle constituent TQ can inhibit CYP2D6 and CYP3A4 and affect the intracellular concentration of drugs metabolized by these enzymes (Al-Jenoobi *et al.*, 2010).

Our data indicated that all tested herbs alter the pharmacokinetic profile of phenytoin albeit to different magnitudes. Garden cress treatment for 7 days caused a modest increase in C_{max} and AUC_{0-t}. These effects were apparently due to reduced phenytoin clearance by ~33%. The elimination half-life, however, did not significantly change, which can be explained by the

simultaneous reduction in tissue distribution at steady state following herb administration in beagle dogs. It is possible that one or more of the garden cress constituents interfere with phenytoin distribution, resulting in a smaller V_{ss}.

The effect of black seed therapy was more significant on phenytoin disposition. The administration of this herbal product caused a significant reduction in the calculated AUC_{0-t} of about 75%. In addition, there was a twofold increase in phenytoin V_{ss}, which might be explained by a possible binding displacement mechanism of this highly plasma protein bound drug (bound fraction ≥0.9) (Rolan, 1994). The active ingredients in the seed include TQ, nigellone and nigellimine, which may dislocate phenytoin from its albumin binding sites and thus generating freer drug to be distributed to the peripheral tissues. Apparently, the magnitude of increased phenytoin clearance in dogs following black seed administration far exceeds that of increased its V_{ss} (eightfold vs. twofold), and therefore, drug elimination half-life was reduced by 77%.

Similarly, fenugreek therapy mimics the effect of black seed, but to a lesser extent. A significant reduction in the calculated AUC_{0-t} of ~50% was observed. A fourfold increase in phenytoin clearance results in a steeper terminal slope compared to the control state. In addition, there was a 73% increase in phenytoin V_{ss}, which is a more reliable estimate of drug's volume of distribution since it is independent of the rate

elimination constant. A possible phenytoin binding displacement effect may also exist in the presence of fenugreek constituents (i.e. alkaloids and flavonoids) in the blood.

The hydroxylation of phenytoin in dog is believed to be mediated by dog orthologues of human CYP2C9 and 2C19 (Graham *et al.*, 2003). Canine's CYP2B11 and 3A12 are possible CYP450 isoforms responsible for phenytoin metabolism in dogs (Nishibe *et al.*, 1998; Graham *et al.*, 2002). Our findings suggest that these herbal products can significantly alter phenytoin pharmacokinetics *in vivo* possibly by modulating the activity of these enzymes and/or altering phenytoin protein binding. Additional studies, however, are warranted to evaluate the molecular mechanisms by

which specific herbal components can affect phenytoin disposition and the clinical consequence of these herb-drug interactions in humans.

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Conflict of Interest

The authors have confirmed that there is no conflict of interest.

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